Chapter 2 - Boolean algebra & Logic gate

Binary logic

1 - Variables: two discrete values

2 - Operations that assume logical meaning. Ex: is equivalent to Bool alg

Boolean algebra

1) Boolean variable: 1 & 0, True & False

2) Boolean operations:

Basic operations: AND, OR, NOT

Basic Boolean operations

AND $x \cdot y$ / xy

x	y	xy
0	0	0
0	1	0
1	0	0
1	1	1

OR $x+y$

x	y	x+y
0	0	0
0	1	1
1	0	1
1	1	1

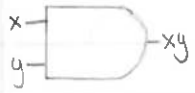
NOT

x	x'
0	1
1	0

Logic gates

Basic

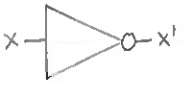
AND



OR



NOT



Boolean expressions (functions)

$f(x, y, z) = xy' + zx'$

name of function | Input variables

term: implementation with a gate

literal: any occurrence of an input variable or its complement.

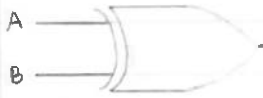
Universal

NAND



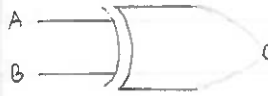
A	B	F
0	0	1
0	1	1
1	0	1
1	1	0

XOR



A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

XNOR



A	B	F
0	0	1
0	1	0
1	0	0
1	1	1

NOR



A	B	F
0	0	1
0	1	0
1	0	0
1	1	0

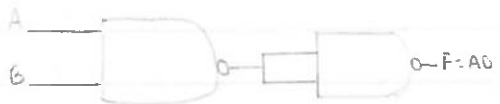
Theorems of Boolean Algebra

1. $X+0=X$
2. $X \cdot 1 = X$
3. $X + X' = 1$
4. $X \cdot X' = 0$
5. $X+X=X$
6. $X \cdot X = X$
7. $X+1=1$
8. $X \cdot 0 = 0$
9. $(X')' = X$
10. $x+y = y+x$ commutative
11. $xy = yx$ commutative
12. $x+(y+z) = (x+y)+z$ associative
13. $x(yz) = (xy)z$ associative
14. $x(x+y) = xy + xz$ distributive
15. $x + yz = (x+y)(x+z)$ distributive
16. $(x+y)' = x'y'$ demorgan
17. $(xy)' = x' + y'$ demorgan
18. $x + xy = x$ absorption
19. $x(x+y) = x$ absorption

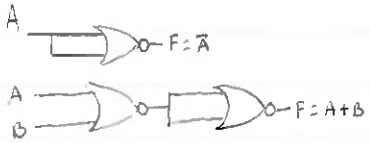
NAND universality

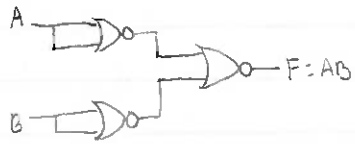


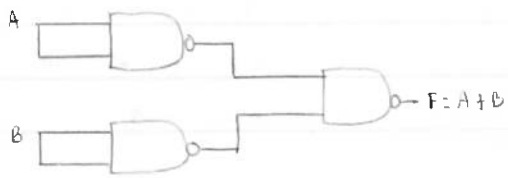
NOT gate



AND Gate







OR gate